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Cutting apparatus

Abstract

A cutting apparatus for a header of an agricultural harvester includes a first wheel and a second wheel. A drive source is configured to selectively drive the first wheel in a clockwise direction or an anticlockwise direction. A knife carrier, including a loop having a running side and a knife-carrying side, is tensioned around the first and second wheels for transmitting torque from the drive source to rotate the wheels and the knife carrier. A plurality of knives cut crop mounted to the knife-carrying surface of the knife carrier. A sensor monitors operation of the knives.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to European Patent Application No. 21180161.8, filed Jun. 17, 2021, the content of such application being incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a cutting apparatus for a header of an agricultural harvester, a header incorporating the cutting apparatus, an agricultural harvester incorporating the header, and to a method of operation of the cutting apparatus.

BACKGROUND

(3) Agricultural harvesters, such as e.g. combine harvesters, include a header that receives, cuts, and gathers crop during harvesting. The header is mounted at the front of the harvester, and is driven towards the crop in order to harvest it. A conventional header includes a frame into which the crop is received, and a sickle cutting bar for cutting the crop ready to be gathered.

(4) The sickle cutting bar comprises a bar from which extends a plurality of cutting knives having a cutting blade on either side. The bar is configured to oscillate a short distance left and right so that the knives are moved leftwards and rightwards in order to provide a cutting action and to cut the crop as it is received into the header.

(5) Unfortunately, the knives are prone to breaking or becoming damaged during operation of the harvester, either due to obstacles in the field colliding with the knives or just due to general wear. Conventionally, the only way to identify a broken knife is the resulting line of unharvested crop extending behind the harvester. The only way to fix the broken knife is to pause harvesting, and replace the knife or, in some cases, the entire sickle bar. This maintenance is far too time-consuming to be practical.

(6) In addition, as the capacity of agricultural harvesters grows and header widths increase, it is becoming more common to drive the harvesters forward faster. The increased forward speed of the harvesters requires a faster cutting oscillation. Moving the knives faster increases the wear on the cutting blades of the knives as well as on the bar itself. Again, there is currently no solution to identifying and rectifying the problem of wear without the costly stoppage of harvesting.

(7) It is an aim of the present invention to address one or more of the disadvantages associated with the prior art.

SUMMARY OF THE INVENTION

(8) According to an aspect of the present invention there is provided a cutting apparatus for a header of an agricultural harvester. The apparatus comprises: a first wheel and a second wheel, both having a drive surface; a drive source configured to selectively drive the first wheel in one of a clockwise direction and an anticlockwise direction; a knife carrier that comprises a loop having a running side and a knife-carrying side, and wherein the loop is tensioned around the first and second wheels so that the running side is engaged by the drive surfaces of the first and second wheels for transmitting torque from the drive source to rotate the wheels and the knife carrier; a plurality of knives for cutting crop, the knives having two cutting blades and being mounted to the knife-carrying surface of the knife carrier; and a sensor for monitoring operation of the knives.

(9) When compared with existing sickle cutting bars, the cutting apparatus described above brings several advantages. One advantage is that the looped knife carrier enables the knives to be moved in a circular motion, so that the knives can be moved continuously in one direction. When the one side of the knives is blunt, the direction can be reversed. By providing a loop, more knives are provided, meaning that the wear on each knife is reduced. Moreover, the loop permits fewer changes of direction because continuous movement in one direction is possible, and this reduces the stresses on the knives and knife carrier. Reducing stresses felt by the knives reduces the risk of knives or of the knife carrier being damaged or broken.

(10) Importantly, because the knife carrier is looped, some knives will be cutting on a cutting side of the apparatus, while other knives will be on the return side of the apparatus and not cutting. The loop therefore permits broken, missing, damaged, or worn knives to be used less and be hidden from the cutting side so that undamaged knives are used to cut the crop. The efficiency of the harvesting can therefore be improved because more of the crop can be cut without pausing to change knives.

(11) In addition, the sensor is important in aiding the control of the knife carrier. By monitoring the operation of the knives, the sensor is able to provide feedback that can be used in controlling the movement of the knife carrier by the drive source, thereby enabling efficient control for optimising the harvest.

(12) The sensor may monitor operation of the knives directly or indirectly. Monitoring the operation of the knives may comprise monitoring the status of the knives. By being suitable for monitoring, the sensor provides output signals based on passive or active measurements that are indicative of the operation of the knives and that can be analysed, using a controller or processor, to determine the operation of the knives, and particularly whether there are knives that are broken, blunt, damaged, or missing.

(13) The sensor may comprise one or more of: an optical sensor; torque sensor; and an inductive sensor.

(14) The first wheel may comprise a first sprocket, the second wheel may comprise a second sprocket, and the loop may comprise a chain. Alternatively, the first wheel may comprise a first pulley, the second wheel may comprise a second pulley, and the knife carrier may comprise a belt.

(15) According to another aspect of the present invention there is provided a header for an agricultural harvester, the header comprising the cutting apparatus described above.

(16) According to another aspect of the present invention there is provided an agricultural harvester comprising the header described above and a controller operatively coupled to the sensor for receiving a signal from the sensor indicative of the operation of the knives and configured to generate a control signal in dependence of the received signal. The controller may be configured to determine one or more missing knives, damaged knives, and/or broken knives based on the received signal. The control signal may be generated based on the determination.

(17) The controller may be operatively coupled to the drive source, and the control signal may comprise a signal for controlling the drive source. Alternatively or additionally, the controller may be operatively coupled to an alert device, and the control signal may comprise a signal for

operating the alert device to alert an operator of the agricultural harvester to the received signal.

(18) According to another aspect of the present invention there is provided a method of operating the cutting apparatus described above, wherein the cutting apparatus comprises a cutting side and a return side. The method comprises: receiving, from the sensor, a signal relating to the operation of the knives; determining, based on the received signal, a section of the knife carrier to spend more time on the return side of the cutting apparatus than on the cutting side; and controlling the drive source to oscillate the movement of the knife carrier through a series of oscillations, the series of oscillations comprising a movement of the knife carrier in the clockwise direction and a movement of the knife carrier in the anticlockwise direction, so that the determined section spends more time on the return side of the cutting apparatus than on the cutting side, and wherein the oscillations are varied based on the determined section.

(19) The knife carrier may move a different distance in the clockwise direction than in the anticlockwise direction in at least a portion of the oscillations.

(20) The method may further comprise identifying broken, missing, or worn knives based on the received signal, and determining the section of the knife carrier based on the identified broken, missing, or worn knives.

(21) The method may further comprise, based on the received signal, identifying the section of the knife carrier carrying the most broken knives, and determining the section of the knife carrier to spend more time on the return side based on the identification of identifying the section of the knife carrier carrying the most broken knives.

(22) Within the scope of this application it is expressly intended that the various aspects, embodiments, examples and alternatives set out in the preceding paragraphs, in the claims and/or in the following description and drawings, and in particular the individual features thereof, may be taken independently or in any combination. That is, all embodiments and/or features of any embodiment can be combined in any way and/or combination, unless such features are incompatible. The applicant reserves the right to change any originally filed claim or file any new claim accordingly, including the right to amend any originally filed claim to depend from and/or incorporate any feature of any other claim although not originally claimed in that manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) One or more embodiments of the invention will now be described, by way of example only, with reference to the accompanying drawings, in which:
- (2) FIG. 1 shows an agricultural harvester in accordance with an embodiment of the invention;
- (3) FIG. 2 shows a cutting apparatus in accordance with an embodiment of the invention;
- (4) FIG. 3 shows a flow chart governing operation of the cutting apparatus of FIG. 2; and
- (5) FIGS. 4a and 4b each show a cutting apparatus with a missing knife.

DETAILED DESCRIPTION

- (6) FIG. 1 shows an agricultural harvester in the form of a combine harvester **10**, which generally includes front and rear ground-engaging wheels **14**, **16**, a header **18**, a feeder **20**, an operator cabin **22**, a threshing and separation system **24**, a cleaning system **26**, a grain tank **28** and an unloading auger **30**. It should be appreciated that while the agricultural harvester is shown as a combine harvester **10**, the agricultural harvester according to the present invention may be embodied by any construction that allows for crop material to be harvested, such as a conventional combine (which does not have a rotor), rotary combine, hybrid combine, chopper harvester, swath mower etc.
- (7) A header **18** is mounted to the front of the combine harvester **10** and includes a cutting apparatus **34** for severing crops from a field during forward motion of the combine. The severed crop is conveyed to the threshing and separating system **24** by a feeder **20**, and through a cleaning

system **26**. In the threshing and separating system **24** and cleaning system, the grain from the severed crop is separated from the 'material other than grain' e.g., straw, leaves, ears and chaff, before. From the cleaning system **26**, the grain is conveyed to a grain tank **28** where it can be discharged from the combine harvester by an unloading tube **30**.

(8) Returning to the header **18**, and more particularly the cutting apparatus **34**, the combine harvester **10** also includes a controller **36** for controlling the cutting apparatus, and an alert device **38** mounted in the operator cabin **22** for alerting an operator based on the operation of the cutting apparatus **34**. These aspects will be discussed in more detail later.

(9) In general, the cutting apparatus **34** is a set of knives arranged in a loop formation so that the cutting apparatus **34** can be driven continuously in either a clockwise or anticlockwise direction to cut the crop received into the header **18**. The cutting apparatus **34** may also be operated through oscillations as required, either to reduce wear to one side of the knives, or to remove certain knives or regions of the apparatus from being used to cut the crop.

(10) FIG. 2 provides a representation of the cutting apparatus **34**. A housing **19** in which the cutting apparatus **34** is provided and which forms part of the header **18** is represented in FIG. 2 by the dotted box.

(11) For the purpose of cutting the crop received by the header, the cutting apparatus **34** comprises a plurality of knives **102** mounted to a knife carrier **104**. Each of the knives **102** shown here is generally triangular, and is mounted to the knife carrier **104** at a mounting side **106**. The other two sides of the knives **102** extend from the knife carrier **104** and each side has a cutting blade **108** for cutting the crop.

(12) The knives **102** are mounted to the knife carrier **104** which is provided as a loop. The loop has an outer, knife-carrying side **110**, from which the knives **102** extend outwardly. The loop also has an inner, running side **112**. The loop, which in this example is provided as a belt, is tensioned around two running wheels, which are in the form of a first pulley **114** and second pulley **116** in FIG. 2. Each of the first and second pulleys **114**, **116** has a drive surface that engages with the running side **112** of the looped knife carrier **104**. Accordingly, torque can be transmitted from the first pulley **114** to the knife carrier **104** when the first pulley **114** is driven, and from the knife carrier **104** to the second pulley **116** to enable smooth rotation of the knife carrier **104**, and therefore the knives **102**.

(13) In other embodiments, the running wheels comprise first and second sprockets and the loop comprises a chain.

(14) The first pulley **114** is driven by a drive source **118**, which may incorporate a DC motor and a gearbox. The drive source **118** is configured to drive the first pulley **114** selectively in a clockwise direction and an anticlockwise direction. The drive source **118** may also include a brake configured to engage to prevent movement of the knife carrier **104**. For example, the brake may be used where an obstacle is encountered to prevent any potential damage to the knives.

(15) A sensor **120** is provided for monitoring operation of the knives **102**. In FIG. 2, the sensor **120** is a torque sensor configured to monitor operation of the drive source. The torque sensor monitors the torque applied by the drive source to the first pulley **114** in order to drive it. If the torque changes during rotation of the knife carrier, this may be an indication of knife performance changing due to e.g. damage or wear, of knives being broken or missing in part of the knife carrier, or of obstacles in the path of the knives.

(16) Although the sensor **120** is here depicted as a torque sensor, the sensor **120** may also or alternatively comprise: an optical sensor; an acoustic sensor; a vibration sensor; a pressure sensor; a capacitance sensor; a power sensor; and/or an inductive sensor arranged to monitor the knives **102** directly. In these embodiments, an optical or inductive sensor is arranged relative to a position that the knives **102** pass during rotation of the knife carrier **104**, so that the wear of the knives **102** or their status, i.e. whether they are broken, missing, or otherwise damaged, can be monitored. For example, the optical sensor may comprise a camera or an optical encoder. An acoustic sensor may

comprise a microphone configured to detect uncharacteristic noise from the knives. Uncharacteristic noises may comprise screeching or scratching noises that have a high content of high frequencies. A vibration sensor may comprise one or more accelerometers positioned on the knife carrier and/or the knives. Broken, damaged, or missing knives will cause uncharacteristic vibration that can be identified based on analysis of a signal from an accelerometer. A pressure sensor may be arranged to determine pressure within a hydraulic drive line if a hydraulic motor is used to drive the knife carrier. A power sensor may be combined with an electric motor to monitor the power provided by the motor to the knife carrier.

(17) When provided in the agricultural harvester **10**, the drive source **118** is configured to be controlled according to control signals received from the controller **36**. The sensor **120** is configured to monitor the operation of the knives **102** and to provide signals indicative of the operation of the knives **102** to the controller **36**.

(18) The controller **36** is a general input-output processing system configured to receive inputs from the sensor **120** and to output control and command signals to the drive source **118**. The controller **36** may be the general controller for the harvester **10** and so may receive inputs and dispatch outputs other than those used for controlling the cutting apparatus **34**. Alternatively, the controller **36** may be a standalone controller.

(19) As will be well understood by the skilled person, the controller **36** may connect to one or more user interfaces to permit input of user controls, as well as output devices to return information to the user. An example of an output device for communicating pertinent information to the user that the controller **36** may be connected to is the alert device **38**. The alert device **38** may be a display and/or audio device provided in the operator cabin **22** configured to provide visual and auditory notification and warning to the operator of the harvester **10** based on determinations made by and inputs received by the controller **36**. The controller **36** may alternatively or additionally be connected to an alert device **38** remote from the harvester **10** via a communications network.

(20) In operation, the cutting apparatus **34** can be considered to have a cutting section **122** and a return section **124**. The cutting section **122** is exposed, in use, to enable the knives **102** to cut crop received into the housing **19** as they pass along the cutting section **122** by movement of the knife carrier **104**. The knives **102** exit the cutting section **122** and pass into the return section **124**, behind and to the sides of the cutting section **122**, where the knives **102** are not exposed, and are housed so that no cutting is possible.

(21) In FIG. **2**, the cutting section **122** is the part not within the dotted box representing the housing **19**. It is noted that the forward direction of travel of the harvester **10** in this configuration would be downwardly relative to the page, in the direction of the arrow **F**. Accordingly, the return section **124** is the part of the cutting apparatus **34** where knives **102** are in the dotted box.

(22) The cutting apparatus **34** is operable in a number of different ways depending upon how the knives **102** are to be handled and upon the operation and state of the knives **102**. For example, in some circumstances, the aim during operation of the cutting apparatus **34** may be to reduce wear on some or all of the knives **102**. In other circumstances, the aim may be to remove from use one or more knives **102** or points where knives are missing by keeping these parts of the knife carrier **104** and the knives **102** thereon within the return section **124**. Particular examples are discussed below with reference to FIGS. **3** to **4b**.

(23) A general method **150** of operation of the cutting apparatus **34** is provided in FIG. **3**. The method **150** may be performed by the controller **36**. The method **150** includes, at step **150**, receiving a signal relating to the operation of the knives **102**. The signal is received from the sensor **120** of the cutting apparatus **34**. The type of signal depends on the type of sensor **120** used. For example, the signal may comprise a torque signal if the sensor is a torque sensor. The torque sensor can be based on a direct force/torque measurement, an indirect measurement, like a strain measurement or on measurement of the acceleration/deceleration based on the change in rotation/speed.

(24) At step **154**, the method **150** includes determining a section of the knife carrier **104** to spend more time on the return side **124** of the cutting apparatus **34** than on the cutting side, based on the received signal from the sensor **120**. In other words, this step **154** is identifying knives **102** that should not be used or should be used for cutting the crop less than the other knives of the cutting apparatus **34**.

(25) There may be a number of reasons for identifying knives **102** for using less and spending more time on the rear section **124**. For example, one or more of the knives of the plurality of knives **102** may have excessive wear on their blades **108** that leads to inefficient cutting when compared to other knives in the plurality of knives **102** carried by the carrier **104**. In another example, the aim may be to balance the wear to all knives **102**, and so knives that have already been used may be identified for less use. In other examples, one or more of the knives may be damaged, broken, or even missing altogether from the sequence of knives. An example of a cutting apparatus **134** with a missing knife **136** is shown in FIG. **4a**. All features other than the missing knife **136** are the same in FIG. **4a** and so have been labelled with the same reference numerals as FIG. **2**. The missing knife **136** in FIG. **4a** is in the cutting section **122**.

(26) To identify a section of the knife carrier **104** that should be moved to and spend more time in the return section **124**, the signal received from the sensor **120** may be interpreted and analysed. The analysis may be used to identify or determine various properties of the knives **102** and cutting apparatus **34**, from which the section of the knife carrier **104** can be determined.

(27) In particular, although not shown in FIG. **3**, the method **150** may comprise identifying broken, missing, or worn knives based on the received signal, and determining the section of the knife carrier **104** based on the identified broken, missing, or worn knives. If more than one part of the knife carrier **104** has broken, worn, or missing knives, the method may comprise identifying the section of the knife carrier **104** carrying the most broken knives, and determining the section of the knife carrier **104** to spend more time on the return side **124** based on the identification of identifying the section of the knife carrier **104** carrying the most broken knives. Accordingly, in both cases, the most effective part of the knife carrier **104** for cutting is identified.

(28) Returning to the method **150** of FIG. **3**, after the section of the knife carrier that is to spend more time on the return side of the cutting apparatus has been determined, step **156** of the method **150** is to control the drive source **118** to move the knife carrier **104** accordingly. In this step, the drive source is controlled to oscillate the movement of the knife carrier through a series of oscillations, the series of oscillations comprising a movement of the knife carrier in the clockwise direction and a movement of the knife carrier in the anticlockwise direction, so that the determined section spends more time on the return side of the cutting apparatus than on the cutting side. The oscillations are varied based on the determined section.

(29) When applying this method to the cutting apparatus **34** shown in FIG. **4a**, the sensor **120** provides torque measurements as the missing knife moves through the cutting section **122**. The torque measurements are indicative of a missing knife **136**, and so the method, at step **154** identifies a missing knife **136**, and determines the section of the knife carrier in which that missing knife **136** is found. This is the section that is to spend more time in the return section of the cutting apparatus **34**. Accordingly, the drive source **118** is controlled to move the knife carrier **104** so that the section with the missing knife **136** is at the return section **124** and is not used in the cutting region **122**, as illustrated in FIG. **4b**. During subsequent operation, the drive source **118** is controlled to oscillate the knife carrier **104** in the clockwise and anticlockwise directions while maintaining the missing knife **136** in the return section **124**, as indicated by the arrow **138** in FIG. **4b**.

(30) In some embodiments of the control method, some or all of the oscillations may be asymmetric oscillations, by which it is meant that the knife carrier moves a different distance in the clockwise direction than in the anticlockwise direction. Asymmetrical oscillations in cutting provide an efficient cutting action while ensuring that the entire usable section of the knife carrier

is used.

(31) In some embodiments, the sensor **120** may monitor the operation of the cutting apparatus in general in order to ensure that the knife carrier, pulleys, and drive source are well maintained and functioning correctly. Changes in torque of the drive source may be an indication of a change in the tension of the knife carrier, and so may indicate a problem that requires rectification. A control method may be provided to identify problems with the knife carrier, pulley, or drive source and to notify the operator of the harvester accordingly.

(32) In some embodiments, the harvester may be provided with a sharpening device and/or a cleaning device for maintaining the knives. The sharpening and/or cleaning devices may be controlled by the controller to sharpen and clean the knives either during operation of the cutting apparatus or during periods when the cutting apparatus is not operating to cut the crop. The sharpening device and/or cleaning device may be positioned to sharpen and clean respectively in the return section of the knife carrier.

(33) It will be appreciated that various changes and modifications can be made to the present invention without departing from the scope of the present application.

Claims

1. A cutting apparatus for a header of an agricultural harvester, the apparatus comprising: a first wheel and a second wheel, each wheel having a drive surface; a drive source configured to selectively drive the first wheel in one of a clockwise direction and an anticlockwise direction; a knife carrier that comprises a loop having a running side and a knife-carrying side, and wherein the loop is tensioned around the first and second wheels so that the running side is engaged by the drive surfaces of the first and second wheels for transmitting torque from the drive source to rotate the wheels and the knife carrier; a plurality of knives for cutting crop, the knives having two cutting blades and being mounted to the knife-carrying side of the knife carrier; a sensor for monitoring operation of the knives; and a controller that is operatively connected to the sensor and the drive source, wherein the controller is configured to: receive, from the sensor, a signal relating to operation of the knives; and determine, based on the received signal, a section of the knife carrier to spend more time on a return side of the cutting apparatus than on a cutting side of the cutting apparatus.
2. The cutting apparatus of claim 1, wherein the sensor comprises one or more of: an optical sensor; a torque sensor; an acoustic sensor; a vibration sensor; and an inductive sensor.
3. The cutting apparatus of claim 1, wherein the first wheel comprises a first sprocket, the second wheel comprises a second sprocket, and the loop comprises a chain.
4. The cutting apparatus of claim 1, wherein the first wheel comprises a first pulley, the second wheel comprises a second pulley, and the loop comprises a belt.
5. A header for an agricultural harvester, the header comprising the cutting apparatus of claim 1.
6. An agricultural harvester comprising: the header of claim 5; and the controller being operatively coupled to the sensor for receiving a signal from the sensor indicative of the operation of the knives and configured to generate a control signal in dependence of the received signal.
7. The agricultural harvester of claim 6, wherein the control signal comprises a signal for controlling the drive source.
8. The agricultural harvester of claim 6, wherein the controller is operatively coupled to an alert device, and wherein the control signal comprises a signal for operating the alert device to alert an operator of the agricultural harvester to the received signal.
9. In a cutting apparatus for a header of an agricultural harvester, the apparatus including (i) a first wheel and a second wheel, each wheel having a drive surface; (ii) a drive source configured to selectively drive the first wheel; (iii) a knife carrier that comprises a loop having a running side and a knife-carrying side, and wherein the loop is tensioned around the first and second wheels so that

the running side is engaged by the drive surfaces of the first and second wheels for transmitting torque from the drive source to rotate the wheels and the knife carrier; (iv) a plurality of knives for cutting crop, the knives having two cutting blades and being mounted to the knife-carrying side of the knife carrier; and (v) a sensor for monitoring operation of the knives, a method of operating the cutting apparatus comprises: receiving, from the sensor, a signal relating to operation of the knives; determining, based on the received signal, a section of the knife carrier to spend more time on a return side of the cutting apparatus than on a cutting side of the cutting apparatus; controlling the drive source to oscillate the movement of the knife carrier through a series of oscillations, the series of oscillations comprising a movement of the knife carrier in a clockwise direction and a movement of the knife carrier in an anticlockwise direction, so that the determined section of the knife carrier spends more time on the return side of the cutting apparatus than on the cutting side.

10. The method of claim 9, wherein, in at least a portion of the oscillations, the knife carrier moves a different distance in the clockwise direction than in the anticlockwise direction.

11. The method of claim 9, further comprising identifying broken, missing, or worn knives based on the received signal, and determining the section of the knife carrier to spend more time on the return side of the cutting apparatus than on the cutting side based on the identified broken, missing, or worn knives.

12. The method of claim 11, further comprising, based on the received signal, identifying the section of the knife carrier carrying the most broken, missing, or worn knives, and determining the section of the knife carrier to spend more time on the return side of the cutting apparatus than on the cutting side based on the identified section of the knife carrier carrying the most broken, missing, or worn knives.

13. A cutting apparatus for a header of an agricultural harvester, the apparatus comprising: a first wheel and a second wheel, each wheel having a drive surface; a drive source configured to selectively drive the first wheel in one of a clockwise direction and an anticlockwise direction; a knife carrier that comprises a loop having a running side and a knife-carrying side, and wherein the loop is tensioned around the first and second wheels so that the running side is engaged by the drive surfaces of the first and second wheels for transmitting torque from the drive source to rotate the wheels and the knife carrier; a plurality of knives for cutting crop, the knives having two cutting blades and being mounted to the knife-carrying side of the knife carrier; and a sensor for monitoring operation of the knives, wherein the sensor is a torque sensor that is configured to monitor a torque applied by the drive source onto the first wheel, wherein changes in the monitored torque are indicative of operation of the knives.

14. The cutting apparatus of claim 1, wherein the controller is further configured to control the drive source to oscillate the movement of the knife carrier through a series of oscillations, the series of oscillations comprising a movement of the knife carrier in a clockwise direction and a movement of the knife carrier in an anticlockwise direction, so that the determined section of the knife carrier spends more time on the return side of the cutting apparatus than on the cutting side.
